# Statistical Mechanics — Formula Sheet

Exam 77802 A, 2026

## Thermodynamics and potentials

### Thermodynamic potentials

$$
\begin{tikzcd}
E(S,V,N) \arrow[r, leftrightarrow, "V \leftrightarrow P"] \arrow[d, leftrightarrow, "S \leftrightarrow T"'] & H(S,P,N) \arrow[d, leftrightarrow, "S \leftrightarrow T"] \\
F(T,V,N) \arrow[r, leftrightarrow, "V \leftrightarrow P"] \arrow[d, leftrightarrow, "N \leftrightarrow \mu"'] & G(T,P,N) \\
\Omega(T,V,\mu) &
\end{tikzcd}
$$

**Legendre transform:**
$p = \frac{\partial f}{\partial x}$
$f^*(p) = px(p) - f(x(p))$

| | |
|---|---|
| $E = TS - PV + \mu N$ | $dE = TdS - PdV + \mu dN$ |
| $H = E + PV = TS + \mu N$ | $dH = TdS + VdP + \mu dN$ |
| $F = E - TS = -PV + \mu N$ | $dF = -SdT - PdV + \mu dN$ |
| $G = E - TS + PV = \mu N$ | $dG = -SdT + VdP + \mu dN$ |
| $\Omega = F - \mu N = -PV$ | $d\Omega = -SdT - PdV - Nd\mu$ |

*$E = TS - PV + \mu N$ is the Euler relation for an extensive homogeneous system.*

**Processes:** isothermal $\mathrm{d}T = 0$; isochoric $\mathrm{d}V = 0$; isobaric $\mathrm{d}P = 0$; adiabatic $\delta Q = 0$; isentropic $\mathrm{d}S = 0$. For a reversible path $\mathrm{d}S = \delta Q_{\mathrm{rev}}/T$; generally $\mathrm{d}S \geq \delta Q/T$. A reversible adiabatic process is isentropic.

## Thermodynamic quantities

$$\beta = \frac{1}{k_{\mathrm{B}}T}, \qquad \frac{\partial \beta}{\partial T} = -\frac{1}{T^2 k_{\mathrm{B}}} = -\beta^2 k_{\mathrm{B}}, \qquad C = \frac{\partial U}{\partial T}.$$

$$J_i \equiv \frac{\partial E(S, x_i)}{\partial x_i}, \qquad \mathrm{d}E = T\,\mathrm{d}S + \sum_i J_i\,\mathrm{d}x_i.$$

$$E(S, x_i) = T(S, x_i)S + \sum_i J_i x_i \qquad S\,\mathrm{d}T + \sum_i x_i\,\mathrm{d}J_i = 0.$$

## Entropy and maximum entropy

**Boltzmann:** for an isolated macrostate with multiplicity $\Omega$,

$$S_{\mathrm{B}} = k_{\mathrm{B}} \ln \Omega, \qquad \frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N}.$$

**Gibbs–Shannon:**

$$S[p] = -k_{\mathrm{B}} \sum_i p_i \ln p_i, \qquad \sum_i p_i = 1.$$

Maximizing $S[p]$ subject to $\langle A_a \rangle = \sum_i p_i A_{a,i}$ gives

$$p_i = \frac{1}{Z} \exp\left(-\sum_a \lambda_a A_{a,i}\right), \qquad Z = \sum_i \exp\left(-\sum_a \lambda_a A_{a,i}\right).$$

Canonical constraint $\langle E \rangle$: $\lambda_E = \beta = 1/(k_{\mathrm{B}}T)$ and $p_i = Z^{-1}\mathrm{e}^{-\beta E_i}$. Equal probabilities apply in the *isolated total system*; tracing out a heat bath produces the Boltzmann factor $\Omega_B(E_{\mathrm{tot}} - E_i) \propto \mathrm{e}^{-\beta E_i}$.

## Transforms and ensemble choice

**Legendre transform:** if $p = f'(x)$, $f^*(p) = px(p) - f(x(p))$. It exchanges a controlled extensive variable with its intensive conjugate.

**Laplace transform:**

$$\mathcal{L}\{g\}(s) = \int_0^\infty \mathrm{e}^{-sx} g(x)\,\mathrm{d}x.$$

Canonical $Z(\beta)$ is the Laplace transform of the density of states: $Z(\beta) = \int \rho(E)\mathrm{e}^{-\beta E}\,\mathrm{d}E$.

***z*-transform / generating function:**

$$\widetilde{a}(z) = \sum_{N\geq 0} a_N z^N, \qquad Z_\Omega(z) = \sum_{N \geq 0} z^N Z_F, \qquad z = \mathrm{e}^{\beta\mu}.$$

Mostly used in the context of cluster expansion.

| fixed variables | ensemble | potential |
|---|---|---|
| $E, V, N$ | microcanonical | $E$ or $S$ |
| $T, V, N$ | canonical | $F$ |
| $T, P, N$ | isothermal–isobaric | $G$ |
| $T, V, \mu$ | grand canonical | $\Omega$ |

## Partition functions and derivatives

**Canonical ensemble** $(T, V, N)$

$$Z_F = \sum_i \mathrm{e}^{-\beta E_i}, \quad F = -k_{\mathrm{B}}T \ln Z_F, \quad U = -\partial_\beta \ln Z_F,$$

$$S = k_{\mathrm{B}}(\ln Z_F + \beta U), \quad P = k_{\mathrm{B}}T\,\partial_V \ln Z_F,$$
$$\mathrm{Var}(E) = \partial_\beta^2 \ln Z_F = k_{\mathrm{B}}T^2 C_V.$$

Classical phase space:

$$Z_F = \frac{1}{N!h^{3N}} \int \prod_{i=1}^N \mathrm{d}^3p_i\,\mathrm{d}^3q_i\,\mathrm{e}^{-\beta H_N}.$$

**Gibbs ensemble** $(T, P, N)$

$$Z_G(\beta, P, N) = \int_0^\infty \mathrm{d}V\,\mathrm{e}^{-\beta PV} Z_F(\beta, V), \qquad G = -k_{\mathrm{B}}T \ln Z_G.$$

Here the effective energy is the enthalpy (Laplace transform of $Z_F$ in $V$)

$$H = E + PV$$
$$\Rightarrow \langle H \rangle = -\partial_\beta \ln Z_G, \qquad \langle V \rangle = -\frac{1}{\beta}\partial_P \ln Z_G,$$
$$S = k_{\mathrm{B}} \ln Z_G + k_{\mathrm{B}}\beta \langle H \rangle.$$

For spins in a magnetic field,

$$B \leftrightarrow -P, \qquad M = \sum_i s_i \leftrightarrow V, \qquad -BM \leftrightarrow PV.$$

**Grand-canonical ensemble** $(T, V, \mu)$

$$Z_\Omega(\beta, V, \mu) = \sum_{N=0}^\infty e^{\beta\mu N} Z_F(\beta, V, N) = \sum_N z^N Z_F(\beta, V, N)$$

The grand potential is

$$\Omega = -\frac{1}{\beta} \ln Z_\Omega = -PV$$

The mean particle number and its fluctuations are

$$\langle N \rangle = \frac{1}{\beta}\partial_\mu \ln Z_\Omega = z\partial_z \ln Z_\Omega,$$

$$\mathrm{Var}(N) = \frac{1}{\beta^2}\partial_\mu^2 \ln Z_\Omega.$$

The entropy is

$$S = k_{\mathrm{B}} \ln Z_\Omega + k_{\mathrm{B}}\beta\left(\langle E \rangle - \mu \langle N \rangle\right).$$

## Ideal gas and equipartition

Thermal wavelength $\lambda_T = h/\sqrt{2\pi m k_{\mathrm{B}}T}$:

$$Z_F = \frac{1}{N!}\left(\frac{V}{\lambda_T^3}\right)^N, \quad PV = Nk_{\mathrm{B}}T, \quad U = \frac{3}{2}Nk_{\mathrm{B}}T.$$

Using Stirling, $\ln N! \simeq N \ln N - N$:

$$F = -Nk_{\mathrm{B}}T\left[\ln\left(\frac{V}{N\lambda_T^3}\right) + 1\right], \quad S = Nk_{\mathrm{B}}\left[\ln\left(\frac{V}{N\lambda_T^3}\right) + \frac{5}{2}\right].$$

Every independent quadratic term in $H$ contributes $\frac{1}{2}k_{\mathrm{B}}T$ to $U$ and $\frac{1}{2}k_{\mathrm{B}}$ to $C_V$.

## Laplace / saddle-point method

For $N \gg 1$, $\mathrm{e}^{-Nf(x)}$ is dominated by the minimum $x_*$ of $f$:

$$I(N) = \int g(x)\mathrm{e}^{-Nf(x)}\,\mathrm{d}x \simeq g(x_*)\mathrm{e}^{-Nf(x_*)}\sqrt{\frac{2\pi}{Nf''(x_*)}}.$$

In $d$ dimensions, with Hessian $H_{ij} = \partial_i \partial_j f(x_*)$,

$$I(N) \simeq g(x_*) e^{-Nf(x_*)} \left(\frac{2\pi}{N}\right)^{d/2} (\det H)^{-1/2}.$$

Width $\Delta x \sim [Nf''(x_*)]^{-1/2}$; **small parameter:** $1/N$ (or $\beta$ at low $T$). Boundary / degenerate minima need a modified expansion.
**Algorithm** (low $T$, central potential). Want

$$I(\beta) = \int_{\mathbb{R}^3} e^{-\beta U(|\mathbf{q}|)} \, d^3 q, \qquad \beta \gg 1,$$

with a minimum of $U$ at $q_*$.

1. **Radial reduction:**
$$I(\beta) = 4\pi \int_0^\infty q^2 e^{-\beta U(q)} \, dq.$$
2. **Taylor of $U$ about $q_*$:**
$$U(q) = U_0 + U_2(q-q_*)^2 + U_3(q-q_*)^3 + U_4(q-q_*)^4 + \cdots,$$
with $U_0 = U(q_*)$, $U_2 = \frac{1}{2}U''(q_*)$, etc.
3. **Saddle:** set $x = q - q_*$; for $\beta \gg 1$ only $x \sim \beta^{-1/2}$ contributes,
$$I(\beta) \approx 4\pi e^{-\beta U_0} \int_{-\infty}^{\infty} (x+q_*)^2 \exp\bigl(-\beta U_2 x^2 - \beta U_3 x^3 - \cdots\bigr) \, dx.$$
4. **Gaussian moments + corrections:**
$$\int_{-\infty}^{\infty} x^n e^{-ax^2} \, dx = \begin{cases} \dfrac{(n-1)!!}{(2a)^{n/2}} \sqrt{\dfrac{\pi}{a}} & n \text{ even}, \\ 0 & n \text{ odd}. \end{cases}$$
5. **Result:**
$$I(\beta) \approx 4\pi^{3/2} e^{-\beta U_0} q_*^2 \xi \bigl(1 + A_2 \xi^2 + A_4 \xi^4 + \cdots\bigr),$$
with $\xi = 1/(\beta\sqrt{U_2})$ and $A_{2,4}$ fixed by higher derivatives of $U$ at $q_*$.

**Stirling:** $N! = \int_0^\infty x^N e^{-x} \, dx$ gives

$$N! \simeq \sqrt{2\pi N}\, (N/e)^N \left(1 + \frac{1}{12N} + \cdots\right).$$

## Interactions, Mean Field and Phase Transitions

### Transfer matrix: periodic 1D Ising model

$$H = -J \sum_{i=1}^{N} s_i s_{i+1} - h \sum_{i=1}^{N} s_i, \qquad s_i = \pm 1,\ s_{N+1} = s_1.$$

Split the field equally between adjacent bonds:

$$T_{ss'} = \exp\left[\beta J s s' + \frac{\beta h}{2}(s+s')\right], \quad T = \begin{pmatrix} e^{\beta J + \beta h} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J - \beta h} \end{pmatrix}.$$

$$Z_N = \operatorname{Tr} T^N = \lambda_+^N + \lambda_-^N,$$

$$\lambda_\pm = e^{\beta J} \cosh(\beta h) \pm \sqrt{e^{2\beta J} \sinh^2(\beta h) + e^{-2\beta J}}.$$

For $N \to \infty$, $f = -k_B T \ln \lambda_+$ and $m = \partial_{\beta h} \ln \lambda_+$. Insert the diagonal spin operator $S = \operatorname{diag}(1, -1)$:

$$\langle s_0 s_r \rangle = \frac{\operatorname{Tr}(S T^r S T^{N-r})}{\operatorname{Tr} T^N}.$$

**At $h = 0$:**

$$\langle s_0 s_r \rangle = \left(\frac{\lambda_-}{\lambda_+}\right)^r + \left(\frac{\lambda_+}{\lambda_-}\right)^{N-r}.$$

At $h = 0$ and $N \to \infty$:

$$\langle s_0 s_r \rangle = (\tanh \beta J)^r = e^{-r/\xi}, \qquad \xi^{-1} = -\ln(\tanh \beta J) = \ln\left|\frac{\lambda_0}{\lambda_1}\right|.$$

A phase transition would require correlations to persist over arbitrarily large distances, i.e. $\xi \to \infty$. In the 1D Ising model, $|\lambda_1| < \lambda_0$ for every $T > 0$, so $\xi$ remains finite and there is no finite-temperature transition. Generally, the subleading eigenvalue sets the slowest-decaying correction and therefore controls long-range correlations.

### Ising mean-field theory

Consider a system with nearest-neighbor interactions,

$$H_{\text{int}} = -J \sum_{\langle i,j \rangle} s_i s_j.$$

Define the fluctuation around the mean magnetization by

$$\delta s_i = s_i - m.$$

Let $d$ be the spatial dimension and $z = 2d$ the coordination number, i.e. the number of nearest neighbors of each site.
**Partition function:**

$$Z_G = \operatorname{tr} \exp\left\{\beta\left(J \sum_{\langle i,j \rangle} s_i s_j + B \sum_i s_i\right)\right\}.$$

Using $s_i = m + \delta s_i$, using $[\cdots] = \delta s_i \delta s_j + m(\delta s_i + \delta s_j), +m^2$

$$Z_G = \operatorname{tr} \exp\left\{\beta J \sum_{\langle i,j \rangle} [\cdots] + \beta B \sum_i \delta s_i + \beta B N m\right\}.$$

**Mean-field assumption:**

$$\delta s_i \delta s_j \approx 0, \qquad s_i s_j \approx m(s_i + s_j) - m^2.$$

**Approximate partition function:** Since

$$\sum_{\langle i,j \rangle} 1 = \frac{Nz}{2},$$

we obtain

$$Z_G^{\mathsf{MF}} = \exp\left(-\frac{\beta J N z m^2}{2}\right) \left[2\cosh\left(\beta B_{\mathsf{MF}}\right)\right]^N,$$

where

$$B_{\mathsf{MF}} = B + Jzm.$$

The self-consistency equation for the magnetization is

$$m = \frac{1}{\beta N} \frac{\partial \ln Z_G}{\partial B} = \tanh\left[\beta(B + Jzm)\right].$$

**Power-laws** At $h = 0$, $T_c = zJ/k_B$. Near $T_c$:

$$m \sim (-t)^{1/2}, \quad \chi \sim |t|^{-1}, \quad m(T_c, h) \sim h^{1/3}, \qquad t = (T - T_c)/T_c.$$

### Hubbard–Stratonovich transformation

For $a > 0$,

$$e^{\frac{a}{2}x^2} = \frac{1}{\sqrt{2\pi a}} \int_{-\infty}^{\infty} d\phi \, \exp\left[-\frac{\phi^2}{2a} + \phi x\right].$$

For a positive matrix $K$,

$$e^{\frac{1}{2}\boldsymbol{s}^T K \boldsymbol{s}} = \frac{1}{\sqrt{\det(2\pi K)}} \int d^n \phi \, e^{-\frac{1}{2}\boldsymbol{\phi}^T K^{-1} \boldsymbol{\phi} + \boldsymbol{\phi}^T \boldsymbol{s}}.$$

The Hubbard–Stratonovich transformation trades a direct quadratic interaction between the microscopic variables for a linear interaction with a new auxiliary field. The new field $\phi$ is continuous and is integrated over all real values.
After the microscopic variables are summed or integrated out, $\phi$ acquires an effective action and describes fluctuations of the order parameter. Approximating the remaining integral by its dominant saddle point fixes $\phi$ to its most probable value and reproduces the mean-field approximation.

### Landau and Ginzburg–Landau theory

Landau theory uses a coarse-grained order parameter $\varphi$ (e.g. magnetization). (After HS, integrating out the microscopic degrees of freedom turns the auxiliary field $\phi$ into this Landau order parameter $\varphi$.) Above $T_c$, symmetry favors $\varphi = 0$; below $T_c$ the system may choose $\varphi \neq 0$. Expand the free-energy density about $\varphi = 0$:

$$f(T, \varphi) = f_0(T) + \frac{1}{2} a(T) \varphi^2 + \frac{1}{4} u \varphi^4 - h\varphi + \cdots,$$

with $u > 0$ and $a(T) \approx a_0(T - T_c) = a_0 T_c \, t$, $t = (T - T_c)/T_c$, $a_0 > 0$. Equivalently,

$$f(\varphi) = f_0 + \frac{a_0 t}{2} \varphi^2 + \frac{u}{4} \varphi^4 - h\varphi + \cdots.$$

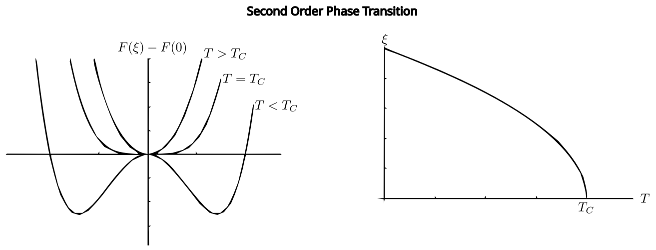
**Minimize:** at $h = 0$,

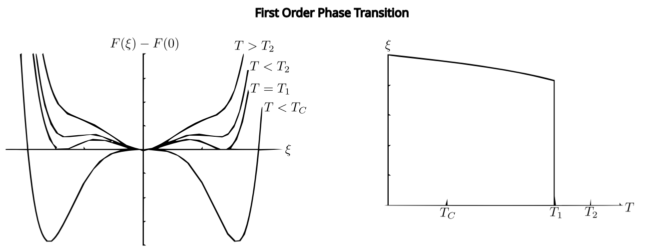
$$\frac{\partial f}{\partial \varphi} = 0 \implies a(T)\varphi + u\varphi^3 = 0.$$

Solutions: $\varphi = 0$ always, and $\varphi = \pm\sqrt{-a(T)/u}$ when $a(T) < 0$ (i.e. $T < T_c$). Near $T_c$,

$$\varphi(T) = \pm\sqrt{\frac{a_0(T_c - T)}{u}} \sim (-t)^{1/2}, \qquad \Rightarrow \quad \beta_{\mathrm{MF}} = \tfrac{1}{2}.$$

If $u > 0$ the order parameter grows continuously from zero (**second-order** / continuous transition). If the quartic coefficient is negative, a stabilizing $+v\varphi^6$ term is needed and the transition is first order; $u = 0$ is the tricritical point.
**Phase classification** (order parameter $\xi$):




Ginzburg–Landau allows $\varphi$ to vary in space:

$$\mathcal{F}[\varphi] = \int \mathrm{d}^d x \left[\frac{c}{2}(\boldsymbol{\nabla}\varphi)^2 + \frac{r}{2}\varphi^2 + \frac{u}{4}\varphi^4 - h\varphi\right], \quad Z = \int \mathcal{D}\varphi\, \mathrm{e}^{-\beta\mathcal{F}[\varphi]}.$$

The gradient penalizes rapid spatial changes; the remaining terms set the locally preferred $\varphi$. Minimizing $\mathcal{F}$ gives

$$-c\nabla^2\varphi + r\varphi + u\varphi^3 = h.$$

Valid near a continuous transition for slow variations on scales $\gg$ lattice spacing (long-wavelength order-parameter fluctuations).

**Critical exponents, scaling and universality**

Near a continuous transition, physical quantities follow power laws:

$$C \sim |t|^{-\alpha}, \quad m \sim (-t)^{\beta}, \quad \chi \sim |t|^{-\gamma}, \quad m(t=0,h) \sim h^{1/\delta},$$

$$\xi \sim |t|^{-\nu}, \qquad G(r, t=0) \sim r^{-(d-2+\eta)}.$$

These powers define the critical exponents. They appear because, as $t \to 0$, the correlation length diverges and the system becomes approximately scale invariant. Widom's scaling hypothesis assumes that the singular part of the free-energy density has the form

$$f_s(t,h) = |t|^{2-\alpha}\mathcal{F}_{\pm}\left(h/|t|^{\beta\delta}\right).$$

Thus $t$ and $h$ do not enter independently, but only through a scale-free combination. Taking derivatives of $f_s$, for example

$$m = -\partial_h f_s, \qquad \chi = \partial_h m,$$

produces the power laws and relations between their exponents:

$$m = |t|^{\beta}\mathcal{M}_{\pm}\left(h/|t|^{\beta\delta}\right),$$

$$\alpha + 2\beta + \gamma = 2, \qquad \gamma = \beta(\delta - 1), \qquad \gamma = \nu(2-\eta), \qquad 2 - \alpha = d\nu \quad .$$

Mean-field values for scalar $\phi^4$ theory are (see 3 boxes above)

$$(\alpha, \beta, \gamma, \delta, \nu, \eta) = (0, \tfrac{1}{2}, 1, 3, \tfrac{1}{2}, 0).$$

Universality means that the exponents depend mainly on spatial dimension, symmetry, number of order-parameter components and interaction range, rather than on microscopic details.

**Fourier-space intuition**

For a lattice field $s_i$ at positions $\boldsymbol{r}_i$,

$$s_i = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} s_{\boldsymbol{k}}\mathrm{e}^{i\boldsymbol{k}\cdot\boldsymbol{r}_i}, \qquad s_{\boldsymbol{k}} = \frac{1}{\sqrt{N}}\sum_i s_i \mathrm{e}^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i}.$$

Translation-invariant quadratic forms diagonalize:

$$\frac{1}{2}\sum_{ij} s_i K_{i-j} s_j = \frac{1}{2}\sum_{\boldsymbol{k}} K(\boldsymbol{k})|s_{\boldsymbol{k}}|^2.$$

Near a transition the soft modes have $ka \ll 1$, so
$K(\boldsymbol{k}) \simeq r + ck^2 + \cdots$; inverse Fourier transformation gives the Ginzburg–Landau gradient term.

## Correlations, Goldstone Modes and Interacting Gases

**Gaussian functional integrals and Green functions**

For a positive operator $K$ and dimensionless source $J$,

$$Z[J] = \int \mathcal{D}\phi\, \exp\left[-\frac{1}{2}\phi K\phi + J\phi\right] = Z[0]\exp\left(\frac{1}{2}JK^{-1}J\right).$$

$$\langle\phi(x)\rangle_J = \frac{\delta \ln Z}{\delta J(x)},$$

$$G_c(x,y) = \langle\phi(x)\phi(y)\rangle_J = \frac{\delta^2 \ln Z}{\delta J(x)\delta J(y)} = K^{-1}(x,y).$$

For the Gaussian Ginzburg–Landau theory,

$$K(\boldsymbol{k}) = r + ck^2, \qquad \widetilde{G}(\boldsymbol{k}) = \frac{1}{r + ck^2}, \qquad \xi = \sqrt{c/r}.$$

At large $R = |\boldsymbol{x}|$,

$$G(R) \sim \frac{\mathrm{e}^{-R/\xi}}{R^{(d-1)/2}} \quad (R \gg \xi), \qquad G(R) \sim R^{-(d-2)} \quad (r = 0,\ d > 2).$$

If the physical field couples as $-h\int\phi$, then

$$\chi = \frac{\partial\langle\phi\rangle}{\partial h} = \beta\int \mathrm{d}^d x\, G_c(x,0) = \beta\widetilde{G}(0).$$

When $J = \beta h$ is used as the source, the same statement is
$\partial\langle\phi\rangle/\partial J = \widetilde{G}(0)$.

**Goldstone modes**

An $O(n)$ field is an $n$-component real field, $\vec{\phi}(\vec{x}) = (\phi_1(\vec{x}), ..., \phi_n(\vec{x}))$, whose free energy is invariant under rotations $R \in O(n)$.
For an $O(n)$ field,

$$\mathcal{F} = \int \mathrm{d}^d x \left[\frac{c}{2}(\boldsymbol{\nabla}\boldsymbol{\phi})^2 + \frac{r}{2}\boldsymbol{\phi}^2 + \frac{u}{4}(\boldsymbol{\phi}^2)^2\right], \qquad r < 0.$$

The minima form a continuous sphere of equivalent ordered states. Spontaneous symmetry breaking means that the system chooses one direction, for example

$$\langle\boldsymbol{\phi}\rangle = v\hat{e}_1, \qquad v^2 = -r/u,$$

and we decompose

$$\boldsymbol{\phi} = (v + \sigma, \pi_1, \ldots, \pi_{n-1}).$$

The longitudinal fluctuation $\sigma$ changes the magnitude of the order parameter and moves away from the minimum, so it costs local energy. The transverse fluctuations $\pi_a$ rotate the order parameter between equivalent minima, so a uniform rotation costs no energy:

$$m_L^2 = 2|r| \quad (\sigma,\text{ longitudinal}), \qquad m_T^2 = 0 \quad (\pi_a,\text{ transverse}).$$

Thus spontaneous breaking $G \to H$ creates $\dim G - \dim H$ massless Goldstone modes: one for each independent direction along the degenerate vacuum manifold. For $O(n) \to O(n-1)$, there are $n - 1$ such modes, with

$$G_T(k) \sim \frac{1}{ck^2}.$$

Only spatial variations of these rotations cost gradient energy, which vanishes for sufficiently long wavelengths.

**Upper/lower critical dimensions and Ginzburg criterion**

The lower critical dimension asks: in how few dimensions can an ordered phase survive fluctuations? For a system with a broken continuous symmetry, the massless Goldstone modes have

$$G_T(k) \sim \frac{1}{k^2}, \qquad \left\langle \pi^2 \right\rangle \sim \int_{k_{\min}} \frac{\mathrm{d}^d k}{k^2}.$$

For $d \leq 2$, the long-wavelength contribution $k \to 0$ diverges. Large-scale rotations then destroy the chosen direction of the order parameter, so there is no finite-$T$ continuous symmetry breaking in short-range systems (Mermin–Wagner). Hence $d_\ell = 2$ for $O(n \geq 2)$. For the discrete Ising symmetry there are no Goldstone modes, and $d_\ell = 1$.
The upper critical dimension asks: above which dimension are fluctuations weak enough that mean-field theory becomes correct? For $\phi^4$,

$$g(\xi) \sim \frac{u k_{\mathrm{B}} T}{c^2}\, \xi^{4-d}$$

measures the strength of fluctuations on the correlation-length scale. Mean field is self-consistent when $g(\xi) \ll 1$ (Ginzburg criterion). Since $\xi \to \infty$ near the transition, fluctuations grow for $d < 4$, are marginal at $d = 4$, and become irrelevant for $d > 4$. Therefore $d_c = 4$: below it the exponents are non-mean-field, at it logarithmic corrections appear, and above it mean-field theory is asymptotically correct. Hyperscaling generally fails above $d_c$.

### Mayer expansion and linked clusters

For pair interactions $U = \sum_{i<j} u(r_{ij})$ define

$$f_{ij} = \mathrm{e}^{-\beta u(r_{ij})} - 1, \qquad \mathrm{e}^{-\beta U} = \prod_{i<j} (1 + f_{ij}).$$

Each term corresponds to a graph: particles are vertices and $f_{ij}$ are edges. The configurational integral is a sum over all graphs. Every graph factorizes into connected components; exponentiating the connected-cluster sum gives

$$\ln Z_\Omega = V \sum_{\ell \geq 1} b_\ell z^\ell, \quad \beta P = \sum_{\ell \geq 1} b_\ell z^\ell, \quad n = \sum_{\ell \geq 1} \ell b_\ell z^\ell,$$

where this convention absorbs the powers of $\lambda_T$ into $z, b_\ell$. The cluster coefficients keep only *connected* graphs $\mathcal{C}_\ell$:

$$b_\ell = \frac{1}{\ell! V} \int_{V^\ell} \left( \sum_{G \in \mathcal{C}_\ell} \prod_{(i,j) \in G} f_{ij} \right) \prod_{i=1}^{\ell} \mathrm{d}^d r_i.$$

The logarithm drops disconnected graphs because they factorize into products already generated by the exponential.
**Example** $\ell = 3$: The connected graphs are the three open chains and the triangle:

$$\sum_{G \in \mathcal{C}_3} \prod f_{ij} = f_{12} f_{13} + f_{12} f_{23} + f_{13} f_{23} + f_{12} f_{13} f_{23}.$$

Hence

$$b_3 = \frac{1}{3! V} \int_{V^3} \left[ f_{12} f_{13} + f_{12} f_{23} + f_{13} f_{23} + f_{12} f_{13} f_{23} \right] \prod_{i=1}^{3} \mathrm{d}^d r_i.$$

The open chains are reducible: removing their central particle disconnects the graph. The triangle remains connected after removing any one particle and is therefore irreducible. Connected graphs determine $\ln Z_\Omega$, while the virial coefficients contain only irreducible graphs; consequently, only the triangle contributes directly to $B_3$. For a partition with $n_k$ clusters of size $k$,

$$\sum_k k n_k = N, \qquad \Omega_N(\{n_k\}) = \frac{N!}{\prod_k (k!)^{n_k} n_k!}.$$

### Virial expansion and basic coefficients

The grand-canonical calculation naturally gives pressure and density as functions of the fugacity $z$. Eliminating $z$ between $P(z)$ and $n(z)$ gives the virial expansion,

$$\beta P = n + B_2 n^2 + B_3 n^3 + \cdots.$$

The first term is the ideal-gas law; the higher powers describe corrections due to interactions between pairs, triples, and larger groups of particles. If

$$\beta P = z + b_2 z^2 + b_3 z^3 + \cdots,$$

then

$$B_2 = -b_2, \qquad B_3 = 4 b_2^2 - 2 b_3.$$

For a translation-invariant pair potential,

$$B_2 = -\frac{1}{2} \int \mathrm{d}^d r\, f(r),$$

$$B_3 = -\frac{1}{3} \int \mathrm{d}^d r_{12}\, \mathrm{d}^d r_{13}\, f_{12} f_{13} f_{23}.$$

Thus $B_2$ measures the effect of one interacting pair, while $B_3$ contains the genuinely three-particle contribution in which all three particles are linked. The virial expansion is useful experimentally because density is measurable, whereas fugacity is not.
For hard spheres of diameter $\sigma$ in three dimensions,

$$B_2 = \frac{2\pi}{3} \sigma^3, \qquad B_3 = \frac{5\pi^2}{18} \sigma^6 = \frac{5}{8} B_2^2.$$

Here $B_2$ is half the volume excluded to the center of a second sphere. More generally, for a $d$-dimensional hard core,

$$B_2 = \tfrac{1}{2} V_d(\sigma), \qquad V_d(R) = \frac{\pi^{d/2} R^d}{\Gamma(d/2 + 1)}.$$

## Stochastic Processes, Noise and Path Integrals

### Markov processes and Chapman–Kolmogorov

A process is Markovian when the future conditioned on the present is independent of the past:

$$P(x_n, t_n | x_{n-1}, t_{n-1}; \ldots; x_0, t_0) = P(x_n, t_n | x_{n-1}, t_{n-1}).$$

Transition kernels obey Chapman–Kolmogorov:

$$P(x_2, t_2 | x_0, t_0) = \int \mathrm{d}x_1\, P(x_2, t_2 | x_1, t_1) P(x_1, t_1 | x_0, t_0).$$

For a homogeneous discrete chain, $\boldsymbol{P}_{n+1} = T \boldsymbol{P}_n$, $\sum_i T_{ij} = 1$, where $T_{ij}$ is the transition probability **from state** $j$ **to state** $i$ ($j \to i$). (Column-stochastic: columns sum to 1.)
A stationary distribution is an eigenvector with eigenvalue 1; modes with $|\lambda| < 1$ decay. Brownian motion is approximately Markovian when molecular collision times are much shorter than the observed time scale.

### Wiener process and Gaussian white noise

A Wiener process $W(t)$ has $W(0) = 0$, continuous paths, independent stationary increments, and

$$W(t + \Delta t) - W(t) \sim \mathcal{N}(0, \Delta t), \quad \langle W(t) \rangle = 0,$$

$$\langle W(t) W(s) \rangle = \min(t, s).$$

Gaussian white noise is the distributional derivative $\eta(t) = \dot{W}(t)$:

$$\langle \eta(t) \rangle = 0, \qquad \langle \eta(t) \eta(t') \rangle = \delta(t - t').$$

For strength $\Lambda$, write $\xi(t) = \sqrt{\Lambda}\, \eta(t)$, so $\langle \xi(t) \xi(t') \rangle = \Lambda \delta(t - t')$. On a grid,

$$\langle \xi_i \xi_j \rangle = \frac{\Lambda}{\Delta t} \delta_{ij}, \qquad \mathcal{P}[\xi] \propto \exp\left[ -\frac{1}{2\Lambda} \int \xi(t)^2\, \mathrm{d}t \right].$$

### Multivariate Gaussian identities

For symmetric positive $A$ and mean zero,

$$p(\boldsymbol{x}) = \frac{\sqrt{\det A}}{(2\pi)^{n/2}} \exp\left( -\frac{1}{2} \boldsymbol{x}^T A \boldsymbol{x} \right), \quad C = A^{-1}.$$

$$I(\boldsymbol{b}) = \left\langle \mathrm{e}^{\boldsymbol{b}^T \boldsymbol{X}} \right\rangle = \exp\left( \frac{1}{2} \boldsymbol{b}^T C \boldsymbol{b} \right), \quad \langle X_i X_j \rangle = C_{ij}.$$

All odd centered moments vanish. Wick/Isserlis theorem:

$$\langle X_1 X_2 X_3 X_4 \rangle = C_{12} C_{34} + C_{13} C_{24} + C_{14} C_{23}.$$

For Gaussian variables, zero covariance implies independence.

### Langevin equations and moments

Underdamped Brownian particle:

$$m \dot{v} = -\gamma v + F(x) + \xi(t), \qquad \langle \xi(t) \xi(t') \rangle = 2 \gamma k_{\mathrm{B}} T \delta(t - t').$$

For $F = 0$, $\tau_B = m/\gamma$ and

$$v(t) = v_0 \mathrm{e}^{-t/\tau_B} + \frac{1}{m} \int_0^t \mathrm{e}^{-(t-s)/\tau_B} \xi(s)\, \mathrm{d}s,$$

$$\langle v\rangle = v_0 e^{-t/\tau_B}, \quad \mathrm{Var}(v) = \frac{k_B T}{m}\left(1 - e^{-2t/\tau_B}\right).$$

Long times: $\langle [x(t) - x(0)]^2\rangle \simeq 2Dt$ in one dimension, $D = k_B T/\gamma$.
Overdamped form:

$$\mathrm{d}x = a(x,t)\,\mathrm{d}t + b(x,t)\,\mathrm{d}W_t.$$

For Itô calculus,

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle A(x)\rangle = \left\langle aA' + \frac{1}{2}b^2 A''\right\rangle.$$

Taking $A = x, x^2, \ldots$ gives moment equations without solving the full PDF.

### Gaussian path integrals

A path integral is an ordinary multidimensional integral with one variable for each time or spatial point:

$$\int \mathcal{D}\phi \sim \lim_{N\to\infty} \int \prod_{i=1}^{N} \mathrm{d}\phi_i.$$

For an ordinary Gaussian integral,

$$\int \mathrm{d}^n x\, e^{-\frac{1}{2}\boldsymbol{x}^T A \boldsymbol{x} + \boldsymbol{J}^T \boldsymbol{x}} = \sqrt{\frac{(2\pi)^n}{\det A}}\, e^{\frac{1}{2}\boldsymbol{J}^T A^{-1}\boldsymbol{J}}.$$

The functional analogue is

$$\int \mathcal{D}\phi\, e^{-\frac{1}{2}\int \phi K \phi + \int J\phi} \propto (\det K)^{-1/2} e^{\frac{1}{2}\int J K^{-1} J}.$$

Thus $K^{-1}$ is the covariance, or Green's function, of the Gaussian field.
For the Langevin equation

$$\dot{x}(t) = a(x,t) + \xi(t), \qquad \langle \xi(t)\xi(t')\rangle = \Lambda\delta(t-t'),$$

a trajectory $x(t)$ requires the noise realization

$$\xi(t) = \dot{x}(t) - a(x,t).$$

Since Gaussian white noise has probability

$$\mathcal{P}[\xi] \propto \exp\left[-\frac{1}{2\Lambda}\int \xi(t)^2\,\mathrm{d}t\right],$$

the corresponding trajectory probability is

$$\mathcal{P}[x] \propto \mathcal{J}[x]\,\exp\left[-\frac{1}{2\Lambda}\int (\dot{x}(t) - a(x,t))^2\,\mathrm{d}t\right].$$

Trajectories close to the deterministic equation

$$\dot{x}(t) = a(x,t)$$

are therefore the most probable.
A functional delta enforces an equation at every time:

$$\delta[F] \propto \int \mathcal{D}\hat{x}\,\exp\left[i\int \hat{x}(t)F(t)\,\mathrm{d}t\right].$$

For example, choosing

$$F(t) = \dot{x}(t) - a(x,t) - \xi(t)$$

imposes the Langevin equation inside the path integral.
The Jacobian $\mathcal{J}[x]$ depends on the discretization convention and is often constant for additive or linear noise.

### Kramers–Moyal expansion

Suppose that during a short time interval $\tau$, a stochastic variable changes by

$$\Delta x = x(t+\tau) - x(t),$$

given that it started at $x(t) = x$.
The short-time motion can be characterized by its moments:

$$\langle \Delta x \mid x(t) = x\rangle, \qquad \left\langle (\Delta x)^2 \mid x(t) = x\right\rangle, \qquad \left\langle (\Delta x)^3 \mid x(t) = x\right\rangle, \ldots$$

The Kramers–Moyal coefficients measure how quickly these moments grow:

$$D^{(n)}(x,t) = \lim_{\tau\to 0}\frac{1}{n!\tau}\langle (\Delta x)^n \mid x(t) = x\rangle.$$

In particular,

$$D^{(1)} = \lim_{\tau\to 0}\frac{\langle \Delta x\rangle}{\tau}$$

is the local average velocity, or drift, while

$$D^{(2)} = \lim_{\tau\to 0}\frac{\langle (\Delta x)^2\rangle}{2\tau}$$

measures the local spreading due to randomness.
These short-time moments determine the evolution of the probability density:

$$\partial_t P(x,t) = \sum_{n=1}^{\infty}(-\partial_x)^n\left[D^{(n)}(x,t)P(x,t)\right].$$

If $D^{(n)} = 0$ for $n \geq 3$, this truncates to the **Fokker–Planck equation**

$$\partial_t P = -\partial_x\left(D^{(1)}P\right) + \partial_x^2\left(D^{(2)}P\right).$$

For the Itô stochastic equation

$$\mathrm{d}x = a(x,t)\,\mathrm{d}t + b(x,t)\,\mathrm{d}W,$$

the short-time change is

$$\Delta x \simeq a\tau + b\Delta W,$$

with

$$\langle \Delta W\rangle = 0, \qquad \left\langle (\Delta W)^2\right\rangle = \tau.$$

Therefore,

$$\langle \Delta x\rangle \simeq a\tau, \qquad \left\langle (\Delta x)^2\right\rangle \simeq b^2\tau,$$

and all higher moments vanish after division by $\tau$ in the limit $\tau \to 0$.
Hence,

$$D^{(1)} = a, \qquad D^{(2)} = \frac{1}{2}b^2, \qquad D^{(n)} = 0 \quad \text{for } n \geq 3.$$

Substituting these coefficients recovers

$$\partial_t P = -\partial_x(aP) + \frac{1}{2}\partial_x^2(b^2 P).$$

Thus, $a$ transports probability, while $b$ causes probability to spread.

## Fokker–Planck, Master Equations and Monte Carlo

### Fokker–Planck equation and probability current

For Itô $\mathrm{d}x = a(x,t)\,\mathrm{d}t + b(x,t)\,\mathrm{d}W_t$,

$$\partial_t P = -\partial_x[aP] + \frac{1}{2}\partial_x^2[b^2 P] = -\partial_x J,$$

$$J = aP - \frac{1}{2}\partial_x(b^2 P).$$

In $d$ dimensions, with $\mathrm{d}x_i = a_i\,\mathrm{d}t + B_{i\alpha}\,\mathrm{d}W_\alpha$ and $D_{ij} = (BB^T)_{ij}$,

$$\partial_t P = -\partial_i(a_i P) + \frac{1}{2}\partial_i\partial_j(D_{ij}P).$$

Boundary conditions: reflecting $J\cdot\hat{n} = 0$; absorbing $P = 0$.
Stationarity requires $\partial_x J = 0$; equilibrium with no net current has $J = 0$. For additive overdamped motion

$$\mathrm{d}x = \frac{F(x)}{\gamma}\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}W, \qquad D = \frac{k_B T}{\gamma},$$

$F = -U'$ gives

$$P_{\mathrm{eq}}(x) = Z^{-1}e^{-\beta U(x)}.$$

### Klein–Kramers equation

For

$$\dot{x} = v, \qquad m\dot{v} = -\gamma v - U'(x) + \xi(t), \qquad \langle \xi(t)\xi(t')\rangle = 2\gamma k_B T\delta(t-t'),$$

$$\partial_t P = -v\partial_x P + \frac{1}{m}U'(x)\partial_v P + \frac{\gamma}{m}\partial_v(vP) + \frac{\gamma k_B T}{m^2}\partial_v^2 P.$$

The stationary solution is Maxwell–Boltzmann:

$$P_{\mathrm{eq}}(x,v) = Z^{-1}\exp\left[-\beta\left(\frac{1}{2}mv^2 + U(x)\right)\right].$$

In the strongly overdamped limit, eliminate $v$ to obtain the Smoluchowski Fokker–Planck equation for $x$.

### Continuous-time master equation

Let $W_{ij}$ be the transition rate $j \to i$. Then

$$\dot{P}_i(t) = \sum_{j \neq i} [W_{ij} P_j - W_{ji} P_i] = \sum_j L_{ij} P_j,$$

where $L_{ij} = W_{ij}$ for $i \neq j$ and $L_{ii} = -\sum_{k \neq i} W_{ki}$, so $\sum_i L_{ij} = 0$. A stationary state satisfies $L\boldsymbol{P}^{\mathrm{st}} = 0$. The zero eigenvalue is the stationary mode; the nonzero eigenvalues determine relaxation times. Multiple closed communicating classes may give multiple stationary states. Probability conservation follows by summing the master equation over $i$.

### Detailed balance

**Equilibrium condition** (pairwise cancellation) w.r.t. $P^{\mathrm{eq}}$:

$$W_{ij} P_j^{\mathrm{eq}} = W_{ji} P_i^{\mathrm{eq}} \qquad \forall i, j.$$

It is sufficient for stationarity but stronger than stationarity: a stationary nonequilibrium state may have circulating probability currents. For the canonical distribution,

$$\frac{W_{ij}}{W_{ji}} = \frac{P_i^{\mathrm{eq}}}{P_j^{\mathrm{eq}}} = \mathrm{e}^{-\beta(E_i - E_j)}.$$

Kolmogorov cycle criterion (for positive rates): detailed balance holds iff the product of rates around every closed cycle equals the product around the reversed cycle.

### Metropolis and Metropolis–Hastings

**Algorithm.** To sample a target $\pi_i$:
1. From state $i$, propose $j$ using $q(i \to j)$.
2. Accept with

$$A(i \to j) = \min\left[1, \frac{\pi_j q(j \to i)}{\pi_i q(i \to j)}\right].$$

   Otherwise remain at $i$.
3. Discard burn-in; estimate observables from the correlated chain and account for autocorrelation.

For **symmetric proposals** ($q(i \to j) = q(j \to i)$) and $\pi_i \propto \mathrm{e}^{-\beta E_i}$,

$$A(i \to j) = \min\left[1, \mathrm{e}^{-\beta(E_j - E_i)}\right].$$

This construction obeys detailed balance. To converge to a unique target, the chain must be irreducible and aperiodic. Smaller proposals accept often but mix slowly; very large proposals are usually rejected.

## Algebra

### Trigonometry

**Angle addition formulas:**

$$\sin(a \pm b) = \sin(a)\cos(b) \pm \cos(a)\sin(b)$$
$$\cos(a \pm b) = \cos(a)\cos(b) \mp \sin(a)\sin(b)$$

**Product-to-sum formulas:**

$$\cos(a)\cos(b) = \frac{1}{2}\left[\cos(a+b) + \cos(a-b)\right]$$
$$\sin(a)\sin(b) = \frac{1}{2}\left[\cos(a-b) - \cos(a+b)\right]$$
$$\cos(a)\sin(b) = \frac{1}{2}\left[\sin(a+b) + \sin(a-b)\right]$$

**Using complex exponentials:**

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i} \qquad \cos(z) = \frac{e^{iz} + e^{-iz}}{2}$$
$$\cosh(x) = \cos(ix) \qquad \sinh(x) = -i\sin(ix)$$

**Double-angle identities:**

$$\cos(2x) = \cos^2(x) - \sin^2(x) \qquad \sin(2x) = 2\sin(x)\cos(x)$$

**Half-angle identities:**

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}} \qquad \cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$
$$\tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \frac{\sin\theta}{1+\cos\theta} = \frac{1-\cos\theta}{\sin\theta}$$

### Powers of Trigonometric Functions

$$\cos^2(x) = \frac{1}{2} + \frac{1}{2}\cos(2x) \qquad \sin^2(x) = \frac{1}{2} - \frac{1}{2}\cos(2x)$$
$$\cos^3(x) = \frac{3}{4}\cos(x) + \frac{1}{4}\cos(3x) \qquad \sin^3(x) = \frac{3}{4}\sin(x) - \frac{1}{4}\sin(3x)$$

### Functional Derivative

For $J[f] = \int_a^b \mathcal{L}\big(x, f(x), f'(x)\big)\,\mathrm{d}x$, we obtain:

$$\frac{\delta J}{\delta f(x)} = \frac{\partial \mathcal{L}}{\partial f}(x) - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial \mathcal{L}}{\partial f'}(x)$$

For $J[f] = \int \mathcal{L}\big(x, f(x)\big)\,\mathrm{d}x$, we obtain:

$$\frac{\delta J}{\delta f(y)} = \frac{\partial \mathcal{L}}{\partial f}\Big(x = y, f(y)\Big)$$

### Integrals

$$\int x e^{ax}\,\mathrm{d}x = \frac{e^{ax}}{a}\left(x - \frac{1}{a}\right) \qquad \int x^2 e^{ax}\,\mathrm{d}x = \frac{e^{ax}}{a}\left(x - \frac{1}{a}\right)$$
$$\int x\cos(ax)\,\mathrm{d}x = \frac{\cos(ax)}{a^2} + \frac{x\sin(ax)}{a}$$
$$\int x\sin(ax)\,\mathrm{d}x = \frac{\sin(ax)}{a^2} - \frac{x\cos(ax)}{a}$$
$$\int x^2\cos(ax)\,\mathrm{d}x = \frac{2x}{a^2}\cos(ax) + \left(\frac{x^2}{a} - \frac{2}{a^3}\right)\sin(ax)$$
$$\int x^2\sin(ax)\,\mathrm{d}x = \frac{2x}{a^2}\sin(ax) + \left(\frac{2}{a^3} - \frac{x^2}{a}\right)\cos(ax)$$
$$\int \sin(ax)\cos(bx)\,\mathrm{d}x = -\frac{\cos[(a-b)x]}{2(a-b)} - \frac{\cos[(a+b)x]}{2(a+b)}$$
$$\int \cos(ax)\cos(bx)\,\mathrm{d}x = \frac{\sin[(a-b)x]}{2(a-b)} + \frac{\sin[(a+b)x]}{2(a+b)}$$
$$\int \sin(ax)\sin(bx)\,\mathrm{d}x = \frac{\sin[(a-b)x]}{2(a-b)} - \frac{\sin[(a+b)x]}{2(a+b)}$$
$$\int \sin^n(ax)\cos(ax)\,\mathrm{d}x = \frac{\sin^{n+1}(ax)}{(n+1)a}$$
$$\int \cos^n(ax)\cos(ax)\,\mathrm{d}x = -\frac{\cos^{n+1}(ax)}{(n+1)a}$$
$$\int e^{ax}\sin(bx)\,\mathrm{d}x = \frac{e^{ax}}{a^2+b^2}\left(a\sin(bx) - b\cos(bx)\right)$$
$$\int e^{ax}\cos(bx)\,\mathrm{d}x = \frac{e^{ax}}{a^2+b^2}\left(a\cos(bx) + b\sin(bx)\right)$$

### Definite Integrals

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\,\mathrm{d}x = \sqrt{\frac{\pi}{\alpha}} \qquad \int_{-\infty}^{\infty} e^{-\alpha x^2 + i\beta x}\,\mathrm{d}x = \sqrt{\frac{\pi}{\alpha}}e^{-\frac{\beta^2}{4\alpha}}$$
$$\int_0^{\infty} e^{-ax}\cos(bx)\,\mathrm{d}x = \frac{a}{a^2+b^2} \qquad \int_0^{\infty} e^{-ax}\sin(bx)\,\mathrm{d}x = \frac{b}{a^2+b^2}$$
$$\int_0^{\pi/2} \sin^2(x)\,\mathrm{d}x = \int_0^{\pi/2} \cos^2(x)\,\mathrm{d}x = \frac{\pi}{4} \qquad \int_{-\pi/2}^{\pi/2} \cos(x)\,\mathrm{d}x = 2$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{a(x)}^{b(x)} f(x)\,\mathrm{d}x = f(b(x))\,b'(x) - f(a(x))\,a'(x)$$

### Basic Differential Equations

$$\ddot{Y} + \lambda^2 Y = 0 \Rightarrow Y = A\sin(\lambda x) + B\cos(\lambda x)$$
$$\ddot{Y} - \lambda^2 Y = 0 \Rightarrow Y = A\sinh(\lambda x) + B\sinh(\lambda x) = Ce^{\lambda x} + De^{-\lambda x}$$
$$\dot{Y} + \lambda Y = 0 \Rightarrow Y = Ae^{\lambda x}$$
$$\ddot{Y} + \lambda = 0 \Rightarrow Y = -\frac{\lambda}{2}x^2 + Ax + B$$
$$r^2 R'' + rR' - \lambda^2 = 0 \Rightarrow R = Ar^{\lambda} + Br^{-\lambda}$$
$$y' + P(x)y = Q(x) \Rightarrow y = e^{-\int P(x)\,\mathrm{d}x}\left(\int e^{\int P(x)\,\mathrm{d}x}Q(x)\,\mathrm{d}x + C\right)$$